Deesara Rupasinghe

Associate Software Engineer | Agentic AI | Python | Full-Stack Development
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PROFESSIONAL SUMMARY

Technology graduate with two Honours bachelors degrees in Data Science and ICT, completed concurrently, demonstrating strong time and workload management. Experienced in contributing to enterprise banking solutions and conducting applied research, with technology projects recognised through national-level awards. Driven by a passion for continuous learning and a humble, growth-oriented mindset, I strive to deliver practical, reliable, and data-driven solutions to real-world problems while continuously improving my technical skills and creating meaningful impact.

TECHNICAL SKILLS

- **Programming Languages:** Python, TypeScript, JavaScript, Java, SQL
- **Software Engineering & Web Development:** React, Node.js, FastAPI, REST APIs, MERN Stack, Spring Boot, Streamlit, ELK Stack
- **Agentic AI & Generative AI:** LLM API Intergration, RAG Systems, AI Agents, Prompt Engineering, Workflow Automation
- **Databases & Management:** PostgreSQL, SQL, Database Design, Data Modelling, Data Warehousing
- **Cloud & DevOps:** AWS, Docker, Git, GitHub, CI/CD
- **Software Testing & Quality Assurance:** Unit Testing, Integration Testing, Debuggins, Error Handling, API Testing
- **Machine Learning:** TensorFlow/Keras, LSTM, Clustering, Predictive Modelling, Model Evaluation & Validation

PROFESSIONAL EXPERIENCE

IFINITY Global - Intern Data Engineer *March 2025 – September 2025*
Key Responsibilities

- Contributed to the development of an enterprise Trade Finance MT/MX solution for the Bank of Ceylon.
- Worked with Oracle Database to design, query, and manage enterprise data.
- Worked with WSO2 products and utilized the ELK Stack for log analysis, monitoring, and data visualization.
- Performed customer segmentation using clustering techniques to support data-driven business insights in data driven dashboard.
- Collaborated with cross-functional teams using Git and Agile development practices.

Technologies: Java, Spring Boot, Oracle, Docker, ELK Stack, Azure AD, SQL, Git

Techwire Lanka (Pvt) Ltd - Intern Software Engineer *March 2021 – June 2021*
Key Responsibilities

- Developed and maintained web applications using C# and the .NET Framework**.**
- Participated in debugging, testing, and code reviews to improve application quality.
- Worked collaboratively with the development team following software engineering best practices.
- Gained practical experience in the complete software development lifecycle (SDLC).

Technologies: C#, .NET Framework, SQL, Git

EDUCATIONAL QUALIFICATIONS

University of Colombo Faculty of Technology Faculty of Technology | 2022 – 2026
Bachelor (Honours) in Information and Communication Technology

Sri Lanka Institute of Information Technology (SLIIT) Faculty of Computing | 2022 – 2026
Bachelor of Science (Honours) in Information Technology (Data Science)

Siri Piyarathana National College Technology Stream | 2018 – 2020
G.C.E Advanced Level Examination – 2 A's & B | Z-Score: 2.27

TECHNICAL PROJECTS

Pawura: Coastal Erosion Prediction & Coral Reef Health Monitoring Final year project UOC – **A+ Grade**
Technologies: Python, Google Earth Engine, TensorFlow/Keras (LSTM), Firebase, ESP32, LoRa, Sentinel-2, Landsat, React, Node.js

- Developed an AI-powered platform to predict coastal erosion and monitor coral reef health.
- Built an LSTM model to forecast shoreline changes using satellite imagery.
- Designed an IoT buoy with environmental sensors for real-time marine data collection.
- Integrated cloud storage and visualization for environmental monitoring.

IoT & AI-Based Decision Support System for HIV Final year project SLIIT– **A Grade**
Technologies: Python, XGBoost, Firebase, PostgreSQL, ESP32, Flutter

- Developed an AI-assisted healthcare decision support system for HIV risk assessment and monitoring with Data warehousing.
- Built machine learning models using healthcare datasets for predictive analysis.
- Integrated wearable IoT sensors for real-time patient monitoring.

Before Buy – AI-Powered Vehicle Emission Analysis Platform
Technologies: Python, AI/LLMs, OCR, Data Analysis, REST APIs, Flutter

- Developing an AI-powered platform to analyze vehicle emission test reports and assess engine condition using historical and recent emission readings.
- Exploring AI-based document understanding to extract and interpret emission report data, addressing OCR accuracy challenges.
- Designing an analysis workflow to identify abnormal emission parameters, compare historical trends, and generate understandable vehicle health insights to support pre-purchase vehicle evaluation.
- Focused on integrating AI-driven analysis into a user-friendly application to simplify vehicle diagnostics for non-technical users.

Construction Safety Monitoring System
Technologies: Arduino, ESP32, Firebase, Machine Learning, React, Node.js, Python

- Developed an IoT-based worker safety monitoring system for construction sites.
- Implemented real-time hazard detection and cloud-based alerts.
- Designed dashboards for monitoring worker safety data.

C Programming Sinhala: Published Android application on Google Play. Link

PUBLICATIONS

Pawura: An Integrated Platform for Coastal Erosion Prediction and Coral Reef Health Monitoring
ICMSS 2026 – International Conference on Mathematical Sciences and Statistics

IoT and AI-Based Decision Support System for HIV: Integrated Risk Assessment, Monitoring, and Treatment Optimization
eSmarTA 2026 – International Conference

AWARDS & ACHIEVEMENTS

Merit Award – NBQSA 2026 (28th National ICT Awards) - Received a Merit Award in the Tertiary Student Project (Technology) category at the 28th National ICT Awards – NBQSA, recognizing innovation and technical excellence in ICT.

1st Runner up at Global Student Entrepreneur Awards (GSEA) 2022/23– Sri Lanka - Recognized as the 1st Runner-up in Sri Lanka's Global Student Entrepreneur Awards for Huwasi Ceylon, a startup focused on exporting premium Ceylon cinnamon to international markets.

Dhack Designathon - 4th Place- Secured 4th Place in the Dhack Designathon by design an innovative technology solution in a competitive UI UX hackathon organized by University of Sri Jayewardenepura

CERTIFICATIONS

- Claude Code in Action - Anthropic
- IBM Data Science Professional Certificate - IBM
- Microsoft Copilot Studio / Agentic AI - TechOne Global
- AWS Academy Cloud Foundations - AWS Academy

REFERENCE - Can be provided upon request.